



Course Title: Pattern Recognition

Course Code: EGCI 463

Undergraduate Program

Mahidol University International College

Division Science

## TQF 3 Course Specifications

### Section 1 General Information

1. Course code and course title

|         |          |                     |
|---------|----------|---------------------|
| Thai    | EGCI 463 | การรู้จำแบบ         |
| English | EGCI 463 | Pattern Recognition |

2. Number of credits 4 (4-0-8)

3. Program and type of subject

|                     |                                                |
|---------------------|------------------------------------------------|
| 3.1 Program         | Bachelor of Engineering (Computer Engineering) |
| 3.2 Type of Subject | Major Course (Required Major)                  |

4. Course Coordinator and Course Lecturer

|     |                    |                               |
|-----|--------------------|-------------------------------|
| 4.1 | Course Coordinator | Lect. Dr. Mingmanas Sivaraksa |
| 4.2 | Course Lecturer    | Lect. Dr. Mingmanas Sivaraksa |

5. Trimester/ Year of Study

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| 5.1 Trimester       | First trimester / for 2 <sup>nd</sup> year Computer Engineering |
| 5.2 Course Capacity | Approximately 20 students                                       |

6. Pre-requisite ICMA 213 Calculus II

7. Co-requisites None

8. Venue of Study Mahidol University, Salaya campus

9. Date of Latest Revision

Date: 13 July 2020



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## Section 2 Goals and Objectives

### 1. Course Goals

After successful completion of this course, students will be able to:

- Describe and analyze the principles of Pattern Recognition
- Choose Appropriate Pattern Recognition techniques
- Apply the knowledge and able to solve various problems using Patter Recognition

### 2. Objectives of Course Development/Revision

#### 2.1 Course Objectives

Students should able to understand and use pattern recognition: features extraction, classification or linear techniques for recognition to solve problems in other field. They should have enough knowledge to pursue new knowledge, develop, choose tools and evaluate the validity of the development using pattern recognition knowledge.

#### 2.2 Course-level Learning Outcomes: CLOs

By the end of the course, students will be able to (CLOs)

CLO1: Use Pattern Recognition techniques to solve problems in other fields

CLO2: Pursue new knowledge from pattern recognition concepts

CLO3: Evaluate the suitability of pattern recognition techniques for each problems

CLO4: Choose pattern recognition tools and techniques for development



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### Section 3 Course Management

#### 1. Course Description

(Thai) การประมวลผลก่อนเพื่อการรู้จำข้อมูล การแยกลักษณะเด่น การจำแนกหรือเทคนิคเชิงเส้นเพื่อการรู้จำ เทคนิคอิงพารามิเตอร์และไร้พารามิเตอร์ ตัวจำแนกแบบเบย์ วิธี โครงข่ายงานระบบประสาท วิธีลูกผสม เทคนิคการเรียนรู้ การประยุกต์การรู้จำ

(English) Preprocessing for data recognition: features extraction, classification or linear techniques for recognition. Parametric and nonparametric techniques, Bayesian classifiers, methods, neural networks, hybrid method, learning techniques. Applications in pattern recognition.

#### 2. Credit hours / trimester

| Lecture<br>(hours)               | Addition<br>al Class<br>(hours) | Laboratory/field<br>trip/internship<br>(hours) | Self-study<br>(hours)            |
|----------------------------------|---------------------------------|------------------------------------------------|----------------------------------|
| 48 hours<br>(4 hours x 12 weeks) | -                               | -                                              | 96 hours<br>(8 hours x 12 weeks) |

#### 3. Numbers of hours that the lecturer provides individual counseling and guidance

1 hour/week



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#### Section 4 Development of Students' Learning Outcome

1. Short summary on the knowledge or skills that the course intends to develop in students (CLOs)

By the end of the course, students will be able to

- CLO1: Use Pattern Recognition techniques to solve problems in other fields
- CLO2: Pursue new knowledge from pattern recognition concepts
- CLO3: Evaluate the suitability of pattern recognition techniques for each problems
- CLO4: Choose pattern recognition tools and techniques for development

2. Teaching methods for developing the knowledge or skills specified in item 1 and evaluation methods of the course learning outcomes

| Course Code | Teaching methods                                                           | Evaluation Methods                                             |
|-------------|----------------------------------------------------------------------------|----------------------------------------------------------------|
| CLO1        | Interactive Lecture, Individual Assignment                                 | Written Examination, Individual Evaluation                     |
| CLO2        | Interactive Lecture, Individual Programming Assignment, Project Assignment | Written Examination, Project Evaluation, Individual Evaluation |
| CLO3        | Interactive Lecture, Case Study, Project Assignment                        | Written Examination, Project Evaluation, Individual Evaluation |
| CLO4        | Interactive Lecture, Case Study, Project Assignment                        | Written Examination, Project Evaluation, Individual Evaluation |



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## Section 5 Teaching and Evaluation Plans

### 1. Teaching plan

| Week  | Topic                                                    | Number of Hours |                                 | Teaching Activities/ Media               | Evaluation                                             |
|-------|----------------------------------------------------------|-----------------|---------------------------------|------------------------------------------|--------------------------------------------------------|
|       |                                                          | Lecture Hours   | Lab/Field Trip/Internship Hours |                                          |                                                        |
| 1     | Introduction and Data preparation for Patter Recognition | 4               | 0                               | Interactive lecture and exercises        | Interactive Lecture                                    |
| 2     | Bayesian Theorem                                         | 4               | 0                               | Interactive lecture and exercises        | Interactive Lecture, Individual Programming Assessment |
| 3     | Preprocessing & Data Preparation                         | 4               | 0                               |                                          |                                                        |
| 4     | Linear Regression and Recognition                        | 4               | 0                               |                                          |                                                        |
| 5     | Evaluation                                               | 4               | 0                               |                                          |                                                        |
| 6     | Applications                                             | 4               | 0                               | Interactive lecture and Group Assignment | Midterm Assessment                                     |
| 7     | Midterm Review & Assessment                              | 2               | -                               |                                          | Written Exam                                           |
| 7-8   | Perceptron                                               | 4               | 0                               | Lecture and homework                     | Interactive Lecture, Individual Programming Assessment |
| 8-9   | Neural Networks                                          | 6               | 0                               | Lecture and homework                     | Interactive Lecture, Individual                        |
| 10    | Dimension Reduction                                      | 2               | 0                               |                                          |                                                        |
| 10-11 | Deep Learning                                            | 4               | 0                               | Lecture and homework                     |                                                        |



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|    |                                  |    |    |                             |                                         |
|----|----------------------------------|----|----|-----------------------------|-----------------------------------------|
| 11 | Unsupervised Learning            | 2  | 0  | Lecture and homework        | Final Assessment                        |
| 12 | Advanced Topics and Applications | 4  | 0  | Discussion and presentation | Interactive Lecture, Project Assessment |
|    | Total                            | 24 | 28 |                             |                                         |

## 2. Plan for Assessing Course Learning Outcomes

### 2.1 Assessing and Evaluating Learning Achievement

#### a. Formative Assessment

The assessment tools such as exercises and assignments are used to evaluate student's understanding by their ability to choose, create and evaluate pattern recognition tools and techniques. Group assignment aims to build skills in pursuing new knowledges.

#### b. Summative Assessment

##### (1) Tools and Percentage Weight in Assessment and Evaluation

| Learning Outcomes                                                                  | Assessment Methods    | Assessment Ratio (Percentage) |    |
|------------------------------------------------------------------------------------|-----------------------|-------------------------------|----|
| CLO1: Use Pattern Recognition techniques to solve problems in other fields         | Individual Assignment | 10                            | 25 |
|                                                                                    | Midterm Examination   | 5                             |    |
|                                                                                    | Final Examination     | 10                            |    |
| CLO2: Pursue new knowledge from pattern recognition concepts                       | Individual Assignment | 10                            | 20 |
|                                                                                    | Midterm Examination   | 5                             |    |
|                                                                                    | Group Project         | 5                             |    |
| CLO3: Evaluate the suitability of pattern recognition techniques for each problems | Individual Assignment | 5                             | 25 |
|                                                                                    | Midterm Examination   | 5                             |    |
|                                                                                    | Final Examination     | 5                             |    |
|                                                                                    | Group Project         | 10                            |    |



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|                                                                       |                       |    |     |
|-----------------------------------------------------------------------|-----------------------|----|-----|
| CLO4: Choose pattern recognition tools and techniques for development | Individual Assignment | 5  | 30  |
|                                                                       | Midterm Examination   | 5  |     |
|                                                                       | Final Examination     | 10 |     |
|                                                                       | Group Project         | 10 |     |
| Total                                                                 |                       |    | 100 |

## (2) Grading System

| Grade | Achievement | Final Score (% range) | GPA |
|-------|-------------|-----------------------|-----|
| A     | Excellent   | 90-100                | 4.0 |
| B+    | Very Good   | 85-89                 | 3.5 |
| B     | Good        | 80-84                 | 3.0 |
| C+    | Fairly Good | 75-79                 | 2.5 |
| C     | Fair        | 70-74                 | 2.0 |
| D+    | Poor        | 65-69                 | 1.5 |
| D     | Very Poor   | 60-64                 | 1.0 |
| F     | Fail        | Less than 60          | 0.0 |

## (3) Re-examination (If course lecturer allows to have re-examination)

N/A - (Not applicable with MUIC)

## 3. Student Appeals

The student wishing to appeal according to grading result must submit a written and signed appeal form personally to the academic affair unit. It is prohibited to assign another person to appeal on one's behalf. The written appeal form is then sent to the program director and chair of department. The final decision is transferred for approval by the faculty committee. The result of appeal then is informed to the student.



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## Section 6 Teaching Materials and Resources

### 1. Textbooks and/or other documents/materials

Nabney, I.T. NETLAB: Algorithms for Pattern Recognition 3<sup>rd</sup> ed: Springer; 2004.

Duda, R.O, Hart, P.E., and Stork, D.G. Pattern Classification, Wiley-interscience, 2001

Bishop, C.M., Pattern Recognition and Machine Learning; Springer; 2007

### 2. Recommended textbooks and/or other documents/materials

### 3. Other Resources (If any)



## Section 7 Evaluation and Improvement of Course Management

1. Strategies for evaluating course effectiveness by students
  - 1.1 Evaluation of peers by students
  - 1.2 Student evaluation
    - 1.2.1 Course content
    - 1.2.2 Course management
    - 1.2.3 Suggestions
    - 1.2.4 Overall opinions
2. Strategies for evaluating teaching methods
  - 2.1 Student evaluation
  - 2.2 Presentation
3. Improvement of teaching methods

Use evaluation from 1 and 2 for course improvement.
4. Verification process for evaluating students' standard achievement outcomes in the course

Analysis of students' learning outcomes using scores from each CLOs for evaluation.
5. Review and plan for improving the effectiveness of the course

Review the course before trimester starts, before each teaching period and review course contents every 3 years.



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## Appendix

### Alignment between Courses and Program

Table 1 The relationship between course and Program Learning Outcomes (PLOs)

|                              | Program Learning Outcomes (PLOs) |      |      |      |      |      |
|------------------------------|----------------------------------|------|------|------|------|------|
|                              | PLO1                             | PLO2 | PLO3 | PLO4 | PLO5 | PLO6 |
| EGCI 465 Pattern Recognition |                                  | R    | R    |      |      | R    |

**Note:** Indicate the level of CLOs by letter I, R, P or M. Using the information as shown in the Curriculum Mapping of TQF2

Table 2 The relationship between CLOs and PLOs

| EGCI 463                                                                           | Program Learning Outcomes (PLOs) |      |      |      |      |      |
|------------------------------------------------------------------------------------|----------------------------------|------|------|------|------|------|
|                                                                                    | PLO1                             | PLO2 | PLO3 | PLO4 | PLO5 | PLO6 |
| CLO1: Use Pattern Recognition techniques to solve problems in other fields         |                                  | 2.1  |      |      |      |      |
| CLO2: Pursue new knowledge from pattern recognition concepts                       |                                  | 2.2  |      |      |      |      |
| CLO3: Evaluate the suitability of pattern recognition techniques for each problems |                                  |      | 3.4  |      |      |      |
| CLO4: Choose pattern recognition tools and techniques for development              |                                  |      |      |      |      | 6.1  |

Table 3 The description of PLOs and Sub Los of the course

| PLOs                                                                                                                         | SubPLOs                                                                      |
|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| PLO2. Integrate computer engineering knowledge with other related sciences and pursue new knowledge in computer engineering. | 2.1 Use computer engineering knowledge to solve problems in other fields     |
|                                                                                                                              | 2.2 Peruse new knowledge in computer engineering using other related science |



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|                                                                                                                                  |                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| PLO 3 Evaluate the computer requirements and identify the appropriate engineering knowledge for developing computer applications | 3.4 Evaluate the suitability of the tools used for different system development        |
| PLO6. Create a related computer engineering development based on information technologies in mathematics or applied statistics.  | 6.1 Choose information technology tools properly for computer engineering development. |